

Hitha Shri Nagaruru

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F-1 student at UCI (MS CS, Dec 2026). CPT-eligible for internship roles. No sponsorship required for internship. Seeking Data Science internship for Summer/Fall 2026.

EDUCATION

University of California, Irvine

Sep 2025 – Dec 2026

Master of Computer Science | GPA: 3.83/4.00

JSS Academy of Technical Education, Bengaluru

Dec 2020 – Jun 2024

BE in Computer Science and Engineering | GPA: 3.81/4.00

EXPERIENCE

Bosch Limited — AI Engineer

Aug 2024 – Aug 2025

Digitization & Industry 4.0 — full-time industry role prior to MS

- Executed a full **EDA pipeline** on diesel pump calibration bench data, identifying inefficiencies that improved productivity by 2.6% and cut cycle time by 9 seconds — avoiding ₹40M (~\$480K) in capital expenditure on new test benches.
- Designed an **end-to-end automated data pipeline** converting machine-generated PDFs into structured Excel datasets using OCR (Tesseract, PyMuPDF) and Pandas, integrated with Power BI for real-time OK/NOK defect tracking.
- Built GPT-4o-powered chatbots using **RAG pipelines** and Azure OpenAI; applied retrieval-based evidence selection and fallback logic to achieve 96% accuracy on known failure cases with 100% safe handling of unseen errors.

PROJECTS

Federated Learning Engine with Privacy Risk Detection

Mar 2026 – Present

- Designing a **federated learning system** simulating decentralized clients with non-IID data distributions and FedAvg-based model aggregation.
- Building a **membership inference attack** using model prediction confidence to quantify training data leakage; integrating differential privacy (gradient clipping, noise injection) and measuring privacy-utility tradeoffs across noise levels.

Validation Framework for LLMs

Jan 2026 – Present

- Built a modular **LLM evaluation framework** with a fail-fast sequential pipeline (toxicity → accuracy), producing calibrated 0–1 PASS/FAIL scores per agent with early termination on unsafe or ungrounded responses.
- Implemented 3-layer toxicity detection: rule-based filtering, **Detoxify scoring**, and SentenceTransformer semantic similarity across policy categories (hate, self-harm, violence) with configurable thresholds.

SKILLS

Languages & Libraries: Python, SQL, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Hugging Face, PySpark

ML & Statistics: Supervised/Unsupervised Learning, Hypothesis Testing, Anomaly Detection, Federated Learning, Differential Privacy, EDA

Data & Visualization: PostgreSQL, Oracle, MySQL, Databricks, Azure ML Studio, Power BI, ETL/ELT

Generative AI: RAG, Prompt Engineering, LLM Evaluation (DeepEval, GEval), Azure OpenAI, Anthropic Claude